

MAX17201/MAX17205/ MAX17211/MAX17215

Stand-Alone ModelGauge m5 Fuel Gauge with SHA-256 Authentication

General Description

The MAX1720x/MAX1721x are ultra-low power stand-alone fuel gauge ICs that implement the Maxim ModelGauge™ m5 algorithm without requiring host interaction for configuration. This feature makes the MAX1720x/MAX1721x excellent pack-side fuel gauges. The MAX17201/MAX17211 monitor a single cell pack. The MAX17205/MAX17215 monitor and balance a 2S or 3S pack or monitor a multiple-series cell pack.

To prevent battery pack cloning, the ICs integrate SHA-256 authentication with a 160-bit secret key. Each IC incorporates a unique 64-bit ID.

The ModelGauge™ m5 algorithm combines the short-term accuracy and linearity of a coulomb counter with the long-term stability of a voltage-based fuel gauge, along with temperature compensation to provide industry-leading fuel gauge accuracy. The IC automatically compensates for cell aging, temperature, and discharge rate, and provides accurate state of charge (SOC) in milliampere-hours (mAh) or percentage (%) over a wide range of operating conditions. As the battery approaches the critical region near empty, the ModelGauge m5 algorithm invokes a special error correction mechanism that eliminates any error. The ICs provide accurate estimation of time-to-empty and time-to-full, Cycle+™ age forecast, and three methods for reporting the age of the battery: reduction in capacity, increase in battery resistance, and cycle odometer.

The ICs provide precision measurements of current, voltage, and temperature. Temperature of the battery pack is measured using an internal temperature measurement and up to two external thermistors supported by ratiometric measurements on auxiliary inputs. A Maxim 1-Wire® (MAX17211/MAX17215) or 2-wire I²C (MAX17201/MAX17205) interface provides access to data and control registers. The ICs are available in lead-free, 3mm x 3mm, 14-pin TDFN and 1.6mm x 2.4mm 15-bump WLP packages.

Applications

- Smartphones and Tablets
- Portable Game Players
- e-Readers
- Digital Still and Video Cameras
- Handheld Computers and Terminals
- Portable Medical Equipment
- Handheld Radios

Benefits and Features

- ModelGauge m5 Algorithm
 - Eliminates Error when Approaching Empty Voltage
 - Eliminates Coulomb-Counter Drift
 - Current, Temperature, and Age Compensated
 - Does Not Require Empty, Full, or Idle States
 - No Characterization Required for EZ Performance (See the [ModelGauge m5 EZ Performance](#) Section)
 - Cycle+ Age Forecasting Observes Lifespan
- Nonvolatile Memory for Stand-Alone Operation
 - Learned Parameters and History Logging
 - Up to 75 Words Available for User Data
- Precision Measurement System
 - No Calibration Required
- Time-to-Empty and Time-to-Full Estimation
- Temperature Measurement
 - Die Temperature
 - Up to Two External Thermistors
- Multiple Series Cell Pack Operation
- Low Quiescent Current
 - MAX172x1: 18µA Active, 9µA Hibernate
 - MAX172x5: 25µA Active, 12µA Hibernate
- Alert Indicator for Voltage, SOC, Temperature, Current, and 1% SOC Change
- High-Speed Overcurrent Comparators
- Predicts Remaining Capacity Under Theoretical Load
- SHA-256 Authentication
- Maxim 1-Wire or 2-Wire (I²C) Interface
- SBS 1.1 Compatible Register Set

[Ordering Information](#) appears at end of data sheet.

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The logo for SKYTECH ELECTRONIC. The word "SKY" is in a large, bold, pink font. The word "TECH" is in a bold, grey font, with the "E" stylized as three horizontal bars. Below "TECH" is the word "ELECTRONIC" in a smaller, bold, grey font.